# Reinforcement Learning with the game Scoundrel

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# Objective

- The original objective was to find an optimized win rate for Scoundrel using reinforcement learning to train an expert agent. This did not work because of the complexity of the observation space and the amount of data needed to play any game of Scoundrel.
- The new objective is being able to find a solution to a given deck of cards. This still requires a large amount of data, but is now within a doable range.

# Reinforcement Learning

To make decisions, we need to make a policy table $\pi(s, a)$ that maximizes our future reward value $r$. We next define a value function that quantifies the desireability of being in a given state:

$$V_\pi(s) = \mathbb{E}\left(\sum_{k=0}^{\infty} \gamma^k r_k | s_0 = s\right)$$

Eventually, we extract the optimal policy as

$$\pi = \arg\max_\pi \mathbb{E}(r_0 + \gamma V(s'))$$

where $s'$ is our next state.

# Proximal Policy Optimization

The goal of PPO is to maximize the following objective function:

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) - c_2 S[\pi_\theta](s_t)]$$

where $c_1$ is the VF coeff., $c_2$ is the entropy coeff., $S$ denotes an entropy bonus, and $L_t^{VF}$ is the squared-error loss.

$$L_t^{VF} = (V_\theta(s_t) - V_t^{targ})^2$$
$$L_t^{CLIP}(\theta) = \hat{\mathbb{E}}\left[\min(r_t(\theta))\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right]$$

# Proximal Policy Optimization

**Algorithm** PPO, Actor-Critic Style

**for** iteration = 1, 2, ... **do**
  **for** actor = 1, 2, ... $N$ **do**
    Run policy $\pi_{\theta_{\text{old}}}$ in environment for $T$ timesteps
    Compute advantage estimates for $\hat{A}_1, \ldots, \hat{A}_T$
  **end for**
  Optimize surrogate $L$ wrt $\theta$ with $K$ epochs and minibatch size $M \leq NT$
  $\theta_{\text{old}} \leftarrow \theta$
**end for**

# Action Space

The model can choose an integer from 0 to 4 to make its choice.

- If it chooses 0, the player runs.
- If it chooses 1-4, then it chooses the card in the position selected.

# Observation Space

The observation space is represented by a vector with a length of 8.

- The first 4 values are the current health, the weapon's value, the current monster on the weapon's value, and whether or not the run button is disabled.
- The last 4 values represent the state of each of the four cards.
  - 0 - There is no card in that slot
  - 1 - There is a monster in the slot
  - 2 - There is a weapon in the slot
  - 3 - There is a health potion in the slot.

# Reward Function

There were many attempts to find the reward function

1. Giving a reward of -1000 on a loss and +1000000 on a win.
2. Giving a reward based on good utilization of the cards. At the end of the turn, it added the health gained and subtracted the damage taken.
3. Giving a reward based on the distance moved towards the goal from before and after the turn. First, the values of all monsters in the deck and hand are added. Then, the player takes their turn. Finally, previous monster count is subtracted by the current monster count and added as a reward.

# Results

```
Trials: 45066  Wins: 406  Score: -26
Trials: 45020  Wins: 400  Score: 0
Trials: 44986  Wins: 429  Score: 3
Trials: 44963  Wins: 430  Score: -5
Trials: 44964  Wins: 430  Score: -23
Trials: 45021  Wins: 401  Score: 3
Trials: 45067  Wins: 407  Score: 3
Trials: 44987  Wins: 429  Score: 0
Trials: 44988  Wins: 429  Score: -61
Trials: 44965  Wins: 431  Score: 3
Trials: 45022  Wins: 401  Score: -7
Trials: 45068  Wins: 408  Score: 3
Trials: 44966  Wins: 431  Score: -140
Trials: 44967  Wins: 431  Score: -105
Trials: 44989  Wins: 429  Score: -26
Trials: 45023  Wins: 402  Score: 3
Trials: 45069  Wins: 408  Score: -26
Trials: 44968  Wins: 432  Score: 3
Trials: 45024  Wins: 402  Score: -9
Trials: 44990  Wins: 429  Score: 0
Trials: 45070  Wins: 409  Score: 3
Trials: 44969  Wins: 433  Score: 3
Trials: 45025  Wins: 402  Score: -46
Trials: 44991  Wins: 429  Score: -26
Trials: 45071  Wins: 410  Score: 3
-----------------------------------------
| rollout/                |           |
|    ep_len_mean          | 60.3      |
|    ep_rew_mean          | 181       |
| time/                   |           |
|    fps                  | 615       |
|    iterations           | 2442      |
|    time_elapsed         | 16261     |
|    total_timesteps      | 10002432  |
| train/                  |           |
|    approx_kl            | 0.0074769985 |
|    clip_fraction        | 0.0766    |
|    clip_range           | 0.2       |
|    entropy_loss         | -0.867    |
|    explained_variance   | 0.97      |
|    learning_rate        | 0.0003    |
|    loss                 | 33.2      |
|    n_updates            | 24410     |
|    policy_gradient_loss | -0.0109   |
|    value_loss           | 53.3      |
-----------------------------------------
```

(a) First Success

```
Trials: 41532  Wins: 24694  Score: 6
Trials: 41529  Wins: 24888  Score: 2
Trials: 41458  Wins: 24973  Score: 10
Trials: 41533  Wins: 24694  Score: -102
Trials: 41530  Wins: 24888  Score: -59
Trials: 41572  Wins: 24649  Score: 6
Trials: 41459  Wins: 24974  Score: 7
Trials: 41534  Wins: 24694  Score: -59
Trials: 41531  Wins: 24889  Score: 10
Trials: 41573  Wins: 24650  Score: 6
Trials: 41460  Wins: 24975  Score: 7
Trials: 41574  Wins: 24650  Score: -177
Trials: 41535  Wins: 24695  Score: 6
Trials: 41532  Wins: 24890  Score: 3
Trials: 41575  Wins: 24651  Score: 7
Trials: 41536  Wins: 24695  Score: -6
Trials: 41461  Wins: 24976  Score: 7
Trials: 41533  Wins: 24890  Score: -51
Trials: 41537  Wins: 24696  Score: 3
Trials: 41576  Wins: 24652  Score: 3
Trials: 41462  Wins: 24977  Score: 9
Trials: 41534  Wins: 24891  Score: 6
Trials: 41577  Wins: 24653  Score: 10
-----------------------------------------
| rollout/                |           |
|    ep_len_mean          | 61        |
|    ep_rew_mean          | 190       |
| time/                   |           |
|    fps                  | 248       |
|    iterations           | 2442      |
|    time_elapsed         | 40324     |
|    total_timesteps      | 10002432  |
| train/                  |           |
|    approx_kl            | 0.013040384 |
|    clip_fraction        | 0.109     |
|    clip_range           | 0.2       |
|    entropy_loss         | -0.93     |
|    explained_variance   | 0.939     |
|    learning_rate        | 0.0003    |
|    loss                 | 24.6      |
|    n_updates            | 24410     |
|    policy_gradient_loss | -0.0133   |
|    value_loss           | 126       |
-----------------------------------------
```

(b) First Test

```
Trials: 43274  Wins: 4450  Score: -5
Trials: 43311  Wins: 4393  Score: -132
Trials: 43330  Wins: 4397  Score: -22
Trials: 43377  Wins: 4342  Score: -39
Trials: 43275  Wins: 4450  Score: -22
Trials: 43312  Wins: 4393  Score: 0
Trials: 43331  Wins: 4397  Score: -22
Trials: 43378  Wins: 4342  Score: -11
Trials: 43313  Wins: 4393  Score: -58
Trials: 43276  Wins: 4451  Score: 3
Trials: 43379  Wins: 4342  Score: -11
Trials: 43314  Wins: 4394  Score: 3
Trials: 43277  Wins: 4452  Score: 2
Trials: 43332  Wins: 4397  Score: -15
Trials: 43380  Wins: 4343  Score: 3
Trials: 43278  Wins: 4452  Score: -17
Trials: 43315  Wins: 4395  Score: 3
Trials: 43333  Wins: 4397  Score: -4
Trials: 43316  Wins: 4396  Score: 3
Trials: 43381  Wins: 4344  Score: 3
Trials: 43279  Wins: 4453  Score: 3
Trials: 43334  Wins: 4397  Score: -8
Trials: 43317  Wins: 4396  Score: -27
Trials: 43280  Wins: 4453  Score: -69
Trials: 43382  Wins: 4344  Score: 0
-----------------------------------------
| rollout/                |           |
|    ep_len_mean          | 66.7      |
|    ep_rew_mean          | 179       |
| time/                   |           |
|    fps                  | 739       |
|    iterations           | 2442      |
|    time_elapsed         | 13523     |
|    total_timesteps      | 10002432  |
| train/                  |           |
|    approx_kl            | 0.012174594 |
|    clip_fraction        | 0.0936    |
|    clip_range           | 0.2       |
|    entropy_loss         | -0.987    |
|    explained_variance   | 0.921     |
|    learning_rate        | 0.0003    |
|    loss                 | 51.5      |
|    n_updates            | 24410     |
|    policy_gradient_loss | -0.0149   |
|    value_loss           | 88.1      |
-----------------------------------------
```

(c) Second Test

```
Trials: 45713  Wins: 1  Score: -8
Trials: 45733  Wins: 1  Score: -36
Trials: 45748  Wins: 0  Score: -11
Trials: 45800  Wins: 0  Score: -9
Trials: 45749  Wins: 0  Score: -85
Trials: 45714  Wins: 1  Score: -10
Trials: 45734  Wins: 1  Score: -8
Trials: 45801  Wins: 0  Score: -8
Trials: 45735  Wins: 1  Score: -20
Trials: 45750  Wins: 0  Score: -9
Trials: 45715  Wins: 1  Score: -8
Trials: 45802  Wins: 0  Score: -9
Trials: 45803  Wins: 0  Score: -170
Trials: 45804  Wins: 0  Score: -123
Trials: 45751  Wins: 0  Score: -8
Trials: 45716  Wins: 1  Score: -8
Trials: 45736  Wins: 1  Score: -8
Trials: 45805  Wins: 0  Score: -9
Trials: 45752  Wins: 0  Score: -11
Trials: 45717  Wins: 1  Score: -8
Trials: 45737  Wins: 1  Score: -8
Trials: 45806  Wins: 0  Score: -11
Trials: 45753  Wins: 0  Score: -8
Trials: 45718  Wins: 1  Score: -8
Trials: 45738  Wins: 1  Score: -8
-----------------------------------------
| rollout/                |           |
|    ep_len_mean          | 58.6      |
|    ep_rew_mean          | 177       |
| time/                   |           |
|    fps                  | 597       |
|    iterations           | 2442      |
|    time_elapsed         | 16743     |
|    total_timesteps      | 10002432  |
| train/                  |           |
|    approx_kl            | 0.015231494 |
|    clip_fraction        | 0.0951    |
|    clip_range           | 0.2       |
|    entropy_loss         | -0.885    |
|    explained_variance   | 0.976     |
|    learning_rate        | 0.0003    |
|    loss                 | 12.7      |
|    n_updates            | 24410     |
|    policy_gradient_loss | -0.0103   |
|    value_loss           | 43.6      |
-----------------------------------------
```

(d) Third Test

# Future Considerations

There are many ways this project could be improved:

- Hyperparameter tuning
- Different algorithms
- Optimizing the observation space
- Interpreting the decisions of the model
- Using exclusively Python

# References I


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# References II


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[5] Antonin Raffin et al. "Stable-Baselines3: Reliable Reinforcement Learning Implementations". In: *Journal of Machine Learning Research* 22.268 (2021), pp. 1–8. URL: http://jmlr.org/papers/v22/20-1364.html.

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# References III

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